

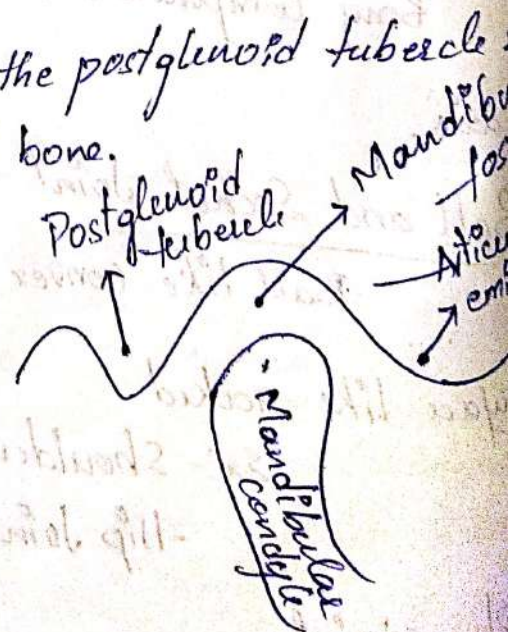
Temporomandibular Joint

- Unique in both structure and function.
- Structurally - Mandibular is a horseshoe-shaped bone.
- Functionally - Mandibular movement involves concurrent movement in the 4 distinct joints, resulting in a complex structure that moves in all planes of motion to achieve normal function.

Joint Structure :-

Articular Structure :-

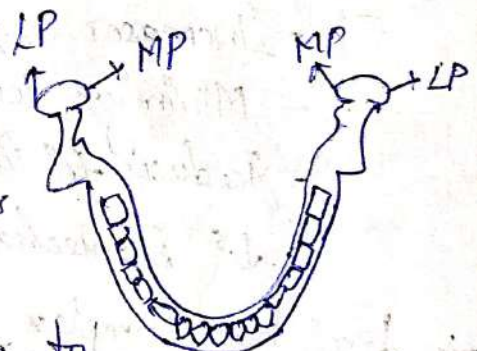
- Multiple bones are merged in TMJ.
 - Mandible
 - Maxillae
 - Temporal
 - Zygomatic
 - Sphenoid
 - Hyoid
- Proximal segment of TMJ is temporal bone.
- Condylus of mandible ~~is~~ sit in mandibular fossa of the temporal bone.
- Mandibular fossa is located b/w the postglenoid tubercle & articular eminence of the temporal bone.



- The articular eminence & the condyle are both convex structure, resulting in an incongruent joint.
- TMJ is classified as a synovial joint, No hyaline cartilage covers the articular surface.
- AE and MC are covered with dense, avascular collagenous tissue that contains cartilaginous cells.
- fibrocartilage is used.
- Fibrocartilage is present because it produces sumodel itself and supai.
- The mandible is the largest of the facial bones & highly mobile (moveable).
- Mandible is arch shape.
- Each mandibular condyle contain medial pole & lateral pole.

→ Lateral pole is readily palpated.

- * External auditory Meatus
- * pushed Anteriorly



→ Coronoid process is located anterior to the mandibular condyle. Attachment to the temporalis muscle.

Accessory Joint Structure :-

- A disc is located within each TM joint.
- It separates the articulation into distinct Superior & Inferior joints.
- Inferior TM Joint - formed with Mandibular condyle & inferior surface of disc.
- Function - Simple hinge joint.

→ Superior TM Joint → formed by articular eminence & Superior surface of disc.

Function - gliding joint.

→ Articular disc is biconcave, Superior & Inferior both are Concave.

→ Thickness of articular disc -

Anteriorly - 2 mm

Posteriorly - 3 mm

Middle - 1 mm

→ Purpose of disc → to allow the convex surface of the articular eminence & mandibular condyle to remain congruent throughout the range of motion of the TMJ in all planes.

- Increases stability

- Minimizes loss of stability

- Reduces friction,

- ↓ biomechanical stress.

→ Disc has complex set of attachment.

→ Disc is attached to medial and lateral poles of the condyle of the mandible.

→ This attachment allow the condyle to rotate freely on the disc in an AP direction.

→ Also attached to joint capsule anteriorly, as well as to the tendon of the lateral Pterygoid muscle.

→ Posteriorly, the disc is attached to bilaminar Retrodiscal pad.

→ Bilaminar Retrodiscal pad contains 2 bands
Superior lamina

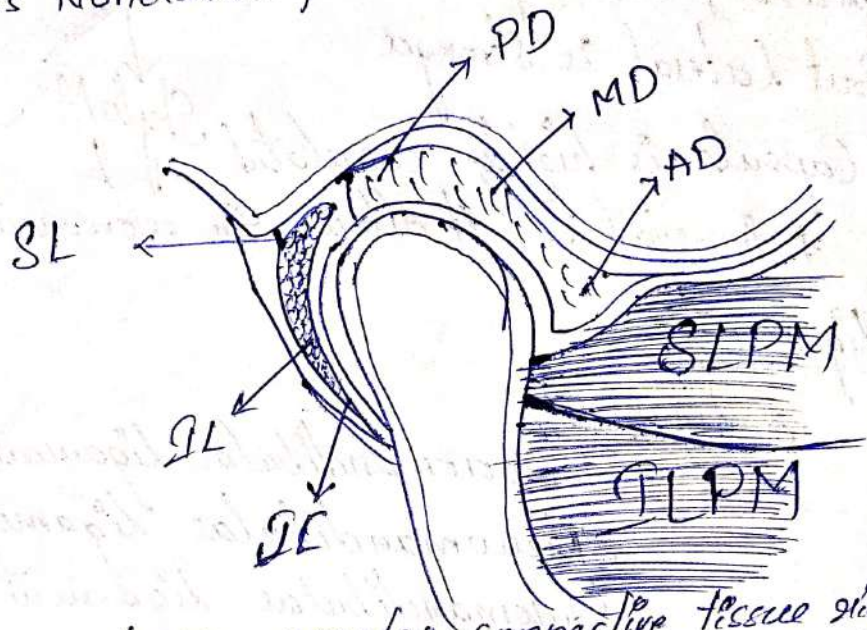
Inferior lamina.

→ Superior lamina - is attached to posterior mandibular fossa.

- It contains elastic fibers and allow to stretch during depression.

→ Inferior lamina - is attached to Neck of the condyle

- It contains Nonelastic fibers.



→ In Both laminae - Loose areolar connective tissue rich in arterial & neural supply is located

Healthy TM disc is viscoelasticity (material resistance to flow) b/w.

Disc contain primarily of collagen, GAGs, elastin (Glycosaminoglycans)

Collagen - maintain disc shape.

Elastin - to Regain its original shape or position.

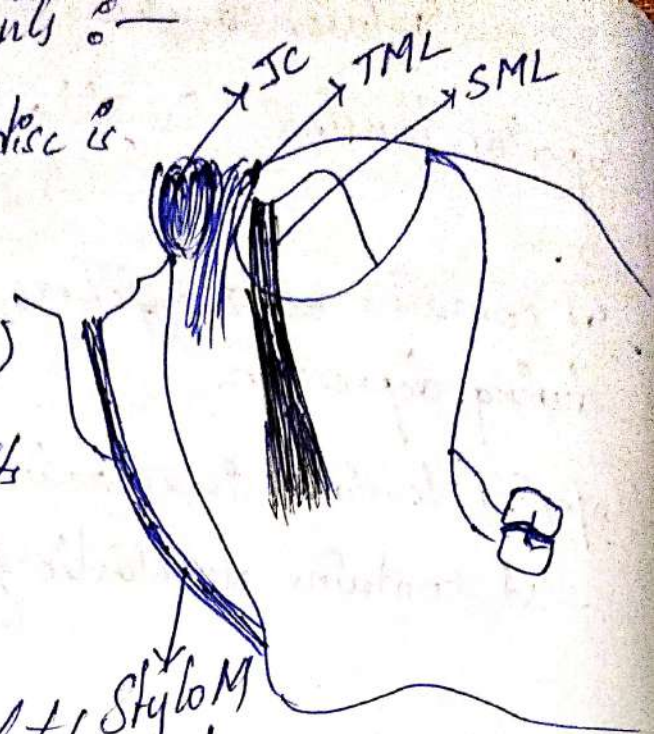
GAGs - Resist mechanical compressive force.

Anterior & posterior disc - Vascular and innervated

Middle disc - Avascular and aneural.

Capsule and Ligaments :-

- Capsule superior to the disc is quite lax. (loose)
- Capsule inferior to the disc is tight. (tight)
- Capsule is thin & loose in its anterior, medial, posterior. But lateral is stronger



- Capsule is highly vasculated & innervated, it allows the movement of the TMJ.

Ligaments -

3 Ligaments.

- Temporomandibular Ligament
- Sphenomandibular Ligament
- Stylomandibular Ligament.

Temporomandibular Ligament →

- It is a strong ligament
- composed 2 parts

① Outer oblique element -

- attached to the neck of the condyle & the articular eminence
- help in mandibular depression

② Inner horizontal element -

- attached to the lateral pole of Condyle & posterior portion of the disc & articular eminence.

Stylomandibular ligament →

- weakest ligament
- attached from the styloid process of the angle to mandible.

Sphenomandibular ligament →

- Strong ligament
- Swinging joint
- Attached from spine of the sphenoid bone to middle surface of ramus of mandible.

Joint Function :-

Joint Kinematics :-

It is the most frequently used mobile joints in the body.
during mastication, swallowing, speaking.

It is a Third order lever.

2 types

Osteokinematic

Arthrokinematic

Osteokinematic - Mandibular - depression, Elevation
- Retraction, Protraction
- Lateral Excursion
Right & Left

Arthrokinematic - Rolling
- Anterior glide
- Lateral glide
- Distraction.

Mandibular Depression & Elevation -

- Fundamental components of mastication.
- Used for gliding & rolling
- Depression ROM - 40 to 50mm
- During mastication - 18mm (depression)
- Rolling initial phase - 11mm or much as 25mm

→ Elevation is reverse to depression.

Mandibular Protrusion and Retrusion -

- Occur in upper TMT.
- Rotation is not present during protrusion & Retrusion
- Teeth are separated during these motion.
- Lower teeth should surpass the upper teeth several millimeter
- Protrusion - necessary for maximal depression.
- Retrusion - m. elevation from a maximally depressed mandible.

Mandibular Lateral Excursion -

- Left & right movement.
- Degree of LE for adult - 8 to 11mm
- Active LE - Contralateral (opp. side)
Ipsilateral (same side)
- Ipsilateral mandibular condyle spins around a vertical axis.

- Contralateral mandibular condyle translates anteriorly along the articular eminence.

Muscles

Primary Muscles

Temporalis
Masseter

Lateral pterygoid
Medial pterygoid

Secondary Muscles

Suprahyoid → Digastric, geniohyoid,
mylohyoid, stylohyoid

Infrahyoid

↓
Omohyoid, Sternohyoid,
Sternothyroid, thyrohyoid.